



BayesianDEB: A Bayesian Framework for Dynamic Energy Budget Modelling in R

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Abstract

Dynamic Energy Budget (DEB) theory provides a mechanistic, thermodynamically consistent description of how organisms acquire and utilise energy throughout their life cycle. The R package **BayesianDEB** embeds the standard DEB model within a Bayesian state-space framework using **Stan** for Hamiltonian Monte Carlo inference. Four pre-written **Stan** programs cover individual growth, growth with reproduction, hierarchical multi-individual analysis, and toxicokinetic-toxicodynamic (TKTD) models for ecotoxicology. A declarative R interface handles the full workflow without requiring **Stan** programming. We validate the individual growth model via simulation-based calibration (500 replications) and show that DEB rate parameters are largely prior-driven under single-individual data, underscoring the importance of species-specific priors or hierarchical pooling. Real-data case studies demonstrate growth estimation and DEBtox EC₅₀/NEC inference. The package is available from <https://sciom.r-universe.dev/BayesianDEB> and <https://github.com/sciom/BayesianDEB>.

Keywords: dynamic energy budget, Bayesian inference, **Stan**, ecotoxicology, DEBtox, TKTD, hierarchical models, R.

1. Introduction

Dynamic Energy Budget (DEB) theory (Kooijman 2010; Nisbet *et al.* 2000; Sousa *et al.* 2008) provides a mechanistic, thermodynamically grounded framework for describing energy acquisition, storage, and allocation across the life cycle of an organism, and is among the most widely applied metabolic theories in ecology and ecotoxicology. Its core state variables — reserve energy E , structural volume V , maturity E_H , and reproduction buffer E_R — evolve according to an ordinary differential equation (ODE) system governed by the κ -rule: a fraction κ of mobilised reserve is allocated to somatic maintenance and growth, while the remainder funds maturity maintenance and reproduction.

Despite its theoretical elegance, DEB calibration has traditionally relied on deterministic least-squares or maximum-likelihood optimisation, as implemented in **DEBtool_M** (Marques *et al.* 2018) and the Add-my-Pet (AmP) online estimation tool. This approach yields point estimates without formal uncertainty quantification, cannot incorporate prior biological knowledge in a principled way, and does not naturally extend to hierarchical data structures where multiple individuals share species-level parameters but differ in, for example, assimilation rate.

The combination of Bayesian inference with ODE-based models is, of course, not new (Gelman *et al.* 2013); Billoir *et al.* (2008) pioneered Bayesian DEBtox calibration using Markov chain Monte Carlo (MCMC), Johnson *et al.* (2015) applied custom samplers to DEB-based thermal-response models, and **deBInfer** (Boersch-Supan *et al.* 2017) provides a general Bayesian ODE framework in R. The contribution of the present work is not a new statistical method but a *domain-specific software system* that makes three concrete advances over existing tools:

1. *Unified Bayesian workflow for DEB.* No existing tool combines DEB ODE models, Hamiltonian Monte Carlo (HMC) inference, observation model switching, Arrhenius temperature correction, convergence diagnostics, posterior predictive checks, derived quantity estimation, and leave-one-out (LOO) cross-validation in a single R interface that requires no recompilation when switching observation families or prior specifications. Users of **DEBtool_M** or custom **Stan** code must assemble this workflow manually; **BayesianDEB** provides it out of the box for the four most common DEB applications.
2. *Reproducible, species-specific prior system grounded in the AmP collection.* The function `prior_species()` returns priors calibrated to AmP parameter estimates (Marques *et al.* 2018) for common test species, bridging the gap between the world’s largest DEB parameter database and Bayesian inference. The prior-to-posterior contraction diagnostic (Section 6.2) quantifies how much the data inform each parameter beyond the prior — a transparency that deterministic calibration cannot offer.
3. *Inference validation for the individual growth model.* Simulation-based calibration (SBC; Section 6.1) with 500 replications and formal χ^2 uniformity tests demonstrates, conditional on MCMC convergence, that the individual growth model is well-calibrated. SBC scripts for the hierarchical, growth-reproduction, and DEBtox models are also provided. This level of algorithmic validation is uncommon in DEB calibration tools.

Together, these advances lower the barrier to rigorous Bayesian DEB modelling from expert **Stan** programming to a declarative R call, while making identifiability limitations and prior sensitivity explicit rather than hidden.

As we show in Section 6.2, growth-only data from a single individual cannot fully resolve the individual DEB rate parameters; informative priors or hierarchical data structures are needed for reliable inference.

The **BayesianDEB** package wraps four **Stan** (Carpenter *et al.* 2017) programs with a declarative R interface, covering the four model types described in Section 2. All models support switchable observation families, Arrhenius temperature correction, threading via `reduce_sum`, and diagnostic tools (Section 4).

This paper describes **BayesianDEB** version 0.1.3. The package is available from the SCIOm r-universe at <https://sciom.r-universe.dev/BayesianDEB> and from GitHub at <https://github.com/sciom/BayesianDEB>.

Table 1: Estimated parameters by model type (I = individual, G = growth_repro, H = hierarchical, D = debtox). State variables (E , V , E_R , D_w) and fixed constants (f , T , T_{ref} , T_A) are described in the text.

Symbol	Description	Units	Models
$\{p_{Am}\}$	Max. assimilation rate	$\text{J d}^{-1} \text{cm}^{-2}$	all
$[p_M]$	Somatic maintenance rate	$\text{J d}^{-1} \text{cm}^{-3}$	all
κ	Allocation fraction	—	all
v	Energy conductance	cm d^{-1}	all
$[E_G]$	Cost of structure	J cm^{-3}	all
E_0, L_0	Initial conditions	J cm^{-3} , cm	all
σ_L	Observation error SD	cm	all
k_J, E_H^p	Maturity maint., puberty	d^{-1} , J	G
k_R, ϕ_R	Repro. conversion, overdisp.	—	G,D
k_d	Damage recovery rate	d^{-1}	D
z_w	No-effect conc. (NEC)	mg kg^{-1}	D
b_w	Effect intensity	$(\text{mg kg}^{-1})^{-1}$	D

2. DEB theory and model formulation

We summarise the DEB equations implemented in **BayesianDEB** and discuss modelling choices; for a full treatment see [Kooijman \(2010\)](#). The standard DEB model is one of several metabolic frameworks; alternatives include the “standard” and “abj” typified models in the AmP collection ([Marques *et al.* 2018](#)), net-production models, and various simplified formulations ([Jager and Zimmer 2012](#)). **BayesianDEB** implements the standard (κ -rule) formulation, which is the most widely used and has the largest empirical parameter base via AmP.

2.1. Standard DEB: growth

Notation. DEB theory uses a specific bracket convention for parameters: curly braces $\{p_{Am}\}$ denote surface-area-specific quantities, square brackets $[p_M]$, $[E_G]$ denote volume-specific quantities, and unbracketed symbols (v , κ) are dimensionless or have other dimensions. We follow this convention throughout.

Table 1 summarises the symbols, units, and roles of all quantities used in this paper.

The two core state variables are reserve energy E (J) and structural volume V (cm^3), with structural length $L = V^{1/3}$. Under constant food availability characterised by the scaled functional response $f \in [0, 1]$, the energy fluxes are:

$$\dot{p}_A = f \{p_{Am}\} L^2, \quad (1)$$

$$\dot{p}_C = \frac{E v L}{E + [E_G] V}, \quad (2)$$

$$\dot{p}_M = [p_M] V, \quad (3)$$

where $\{p_{Am}\}$ is the surface-area-specific maximum assimilation rate ($\text{J d}^{-1} \text{cm}^{-2}$), v is the energy conductance (cm d^{-1}), $[E_G]$ is the specific cost of structure (J cm^{-3}), and $[p_M]$ is the volume-specific somatic maintenance rate ($\text{J d}^{-1} \text{cm}^{-3}$). Note that the mobilisation flux $\dot{p}_C$

(Equation 2) is the total catabolic power drawn from reserve; it is subsequently partitioned by the κ -rule in the state equations below. The state equations are:

$$\frac{dE}{dt} = \dot{p}_A - \dot{p}_C, \quad (4)$$

$$\frac{dV}{dt} = \frac{\kappa \dot{p}_C - \dot{p}_M}{[E_G]}, \quad (5)$$

where κ is the allocation fraction to soma. At steady state ($dV/dt = 0$, $dE/dt = 0$), the ultimate structural length is $L_\infty = f \kappa \{p_{Am}\} / [p_M]$.

Edge cases. When $\kappa \dot{p}_C < \dot{p}_M$ (starvation), growth ceases and the organism draws on structure for maintenance; in our implementation dV/dt is clamped to zero to prevent negative structural volume. When $\kappa \rightarrow 0$, nearly all mobilised energy goes to reproduction and growth is negligible; when $\kappa \rightarrow 1$, reproduction ceases. The Beta prior on κ (default Beta(2, 2)) regularises the posterior away from these boundary cases.

Initial conditions. Both E_0 (initial reserve density, J cm^{-3}) and L_0 (initial structural length) are estimated with log-normal priors. The total initial reserve is $E(0) = E_0 L_0^3$. Estimating E_0 freely avoids assuming energy steady state, which may not hold for hatchlings. In the hierarchical model, E_0 is shared across individuals while $L_{0,j}$ varies.

ODE stiffness and solver choice. DEB systems are mildly stiff due to the time-scale separation between fast reserve dynamics and slow structural growth. The mobilisation flux $\dot{p}_C = E v L / (E + [E_G] V)$ creates stiffness when $E \gg [E_G] V$ (near steady-state reserve). We use the BDF (backward differentiation formula) solver `ode_bdf_tol` with tolerances 10^{-6} , which is standard for mildly stiff biological ODE systems. The explicit Runge-Kutta solver (`ode_rk45`) was also tested: it produced identical trajectories for well-conditioned parameter regions but required approximately $2\times$ more function evaluations and occasionally failed for extreme parameter combinations during MCMC warmup. The BDF solver was therefore chosen as the safer default.

2.2. Growth with reproduction

For post-pubertal organisms ($E_H \geq E_H^p$), adding the reproduction buffer E_R (J) yields a 3-state model:

$$\frac{dE_R}{dt} = (1 - \kappa) \dot{p}_C - k_J E_H^p, \quad (6)$$

where k_J is the maturity maintenance rate coefficient and E_H^p is maturity at puberty (treated as a known constant, not a state variable, since the model describes post-pubertal growth only). Offspring counts over an interval $[t_s, t_e]$ are modelled as $R \sim \text{NegBin}_2(\mu = k_R \Delta E_R, \phi)$, where $\Delta E_R = E_R(t_e) - E_R(t_s)$, k_R is the buffer-to-offspring conversion rate (offspring J^{-1}), μ is the expected count, and $\phi > 0$ is the overdispersion parameter. This uses Stan's `neg_binomial_2` parametrisation, where $E[R] = \mu$ and $\text{Var}[R] = \mu + \mu^2/\phi$; as $\phi \rightarrow \infty$ the distribution converges to $\text{Poisson}(\mu)$.

2.3. DEBtox: toxicokinetic-toxicodynamic extension

The DEBtox framework (Jager *et al.* 2006; Jager and Zimmer 2012), part of the broader toxicokinetic-toxicodynamic (TKTD) family that includes the General Unified Threshold model of Survival (GUTS) for survival (Jager *et al.* 2011), adds a scaled internal damage variable D_w :

$$\frac{dD_w}{dt} = k_d(\max(C_w - z_w, 0) - D_w), \quad (7)$$

where k_d is the damage recovery rate, C_w is the external concentration, and z_w is the no-effect concentration (NEC). The damage causes a stress factor $s = b_w D_w$ that reduces assimilation: $\dot{p}_A \leftarrow \dot{p}_A \cdot \max(1 - s, 0)$. The reproduction buffer in the DEBtox model follows a simplified form $dE_R/dt = (1 - \kappa)\dot{p}_C$ (omitting the maturity maintenance term $k_J E_H^p$ from Equation 6), since the stress on assimilation already captures the dominant toxicant effect on reproduction; consequently, k_J and E_H^p are not parameters of the DEBtox model. At steady state, $D_w = C_w - z_w$ for $C_w > z_w$, so the EC₅₀ for 50% assimilation reduction is:

$$\text{EC}_{50} = z_w + \frac{0.5}{b_w}. \quad (8)$$

2.4. Arrhenius temperature correction

All rate parameters scale with temperature by the factor $c_T = \exp(T_A/T_{\text{ref}} - T_A/T)$ (Kooijman 2010, Eq. 1.2), where T_A is the Arrhenius temperature (species-specific, typically 6000–12 000 K). In **BayesianDEB**, the correction is applied inside the Stan ODE to $\{p_{Am}\}$, $[p_M]$, v , k_J , and k_d ; the estimated parameters correspond to the reference temperature T_{ref} .

2.5. Hierarchical structure

When data from multiple individuals are available, a lognormal random effect on $\{p_{Am}\}$ is introduced:

$$\log\{p_{Am}\}_j = \mu_{\log\{p_{Am}\}} + \sigma_{\log\{p_{Am}\}} \cdot z_j, \quad z_j \sim \mathcal{N}(0, 1), \quad (9)$$

using a non-centred parameterisation to avoid pathological funnel geometry (Betancourt and Girolami 2015). The shared parameters $[p_M]$, κ , v , $[E_G]$ are estimated from all individuals jointly (partial pooling).

Why random effects on $\{p_{Am}\}$ only? In DEB theory, $\{p_{Am}\}$ is the parameter most strongly linked to individual phenotypic variation: it determines food uptake capacity and scales with structural surface area, which varies among individuals due to differences in condition, microhabitat, and genotype. The energy conductance v , maintenance $[p_M]$, and costs $[E_G]$ are typically more conserved within a species (Kooijman 2010; Marques *et al.* 2018). We acknowledge that individual variation in κ and L_0 is also documented empirically; however, placing random effects on all parameters simultaneously would create a high-dimensional, poorly identified model for typical DEB dataset sizes (5–30 individuals with 5–15 time points each). Our contraction analysis (Section 6.2) shows that even shared parameters are weakly identified from growth-only data; adding per-individual variation on multiple parameters would exacerbate this. The single-random-effect design is therefore a deliberate parsimony choice that provides partial pooling on the most variable parameter while keeping the model identifiable. Extension to multivariate random effects on $(\{p_{Am}\}, \kappa)$

or $(\{p_{Am}\}, L_0)$, potentially with informative LKJ priors on the correlation matrix, is planned for a future version.

3. Bayesian state-space formulation

We now formalise the statistical model. Let $\mathbf{x}(t) = (E(t), V(t))^\top$ denote the latent DEB state (extended to $(E, V, E_R)^\top$ or $(E, V, E_R, D_w)^\top$ for the reproduction and DEBtox models), and let $\boldsymbol{\theta}$ collect the DEB parameters.

3.1. Process model

The latent state evolves deterministically according to the ODE:

$$\frac{d\mathbf{x}}{dt} = \mathbf{f}(\mathbf{x}, \boldsymbol{\theta}, f, C_w), \quad \mathbf{x}(0) = \mathbf{x}_0(\boldsymbol{\theta}), \quad (10)$$

where $\mathbf{f}$ is defined by Equations (4)–(5) (growth), (6) (reproduction), or (7) (DEBtox), and $\mathbf{x}_0$ depends on initial conditions (E_0, L_0) .

3.2. Observation model

Given the ODE solution $\hat{L}_i = [\mathbf{x}(t_i)]_V^{1/3}$, the growth observations are conditionally independent:

$$L_i^{\text{obs}} \mid \hat{L}_i, \sigma_L \sim \begin{cases} \mathcal{N}(\hat{L}_i, \sigma_L) & \text{(Gaussian),} \\ \text{LogNormal}(\log \hat{L}_i, \sigma_L) & \text{(log-normal),} \\ t_\nu(\hat{L}_i, \sigma_L) & \text{(Student-}t\text{).} \end{cases} \quad (11)$$

For reproduction, interval offspring counts use Stan’s mean-overdispersion parametrisation (`neg_binomial_2`):

$$R_k \mid \mu_k, \phi \sim \text{NegBin}_2(\mu_k, \phi) \quad \text{or} \quad \text{Poisson}(\mu_k), \quad \mu_k = k_R \Delta E_{R,k}, \quad (12)$$

where $E[R_k] = \mu_k$ and $\text{Var}[R_k] = \mu_k + \mu_k^2/\phi$.

3.3. Prior model

The full prior factorises as:

$$p(\boldsymbol{\theta}) = p(\{p_{Am}\}) p([p_M]) p(\kappa) p(v) p([E_G]) p(E_0) p(L_0) p(\sigma_L) [\text{model-specific}]. \quad (13)$$

Default prior families are: $\{p_{Am}\}, [p_M], v, [E_G] \sim \text{LogNormal}$; $\kappa \sim \text{Beta}$; $\sigma_L \sim \text{HalfNormal}$ (following the recommendations of Gelman 2006). Hyperparameters are calibrated against the AmP collection ranges (Marques *et al.* 2018) and passed to Stan as data, so that changing priors does not require recompilation.

Prior calibration rationale. Table 2 shows the generic default hyperparameters and their implied 95% prior mass ranges. These defaults (from `prior_default()`) are deliberately wide ($\sigma = 1.0$) to accommodate any species; they are *not* suitable for quantitative inference without tightening (see Section 6.2).

For common test species, the function `prior_species()` returns priors calibrated to the AmP collection (Marques *et al.* 2018), with $\sigma = 0.3$ for well-characterised parameters:

Table 2: Default prior specifications and implied ranges, compared with AmP empirical ranges for earthworms and springtails.

Parameter	Prior family	μ / a	σ / b	95% prior range
$\{p_{Am}\}$	LogNormal	1.0	1.0	0.4–20
$[p_M]$	LogNormal	−1.0	1.0	0.04–2
κ	Beta	2	2	0.09–0.91
v	LogNormal	−1.5	1.0	0.02–1
$[E_G]$	LogNormal	6.0	1.0	100–3000
σ_L	HalfNormal	0	0.1	0–0.2

```
R> prior_species("Eisenia_fetida")
```

```
p_Am:  LogNormal(1.61, 0.3)
p_M:   LogNormal(-0.69, 0.3)
kappa: Beta(5, 2)
v:     LogNormal(-3.91, 0.5)
E_G:   LogNormal(8.39, 0.3)
sigma_L: HalfNormal(0.05)
```

```
R> prior_species("Daphnia_magna", type = "debtox")
```

Built-in species include *Eisenia fetida*, *E. andrei*, *Folsomia candida*, *Daphnia magna*, and *Lumbricus rubellus*. Users working with other species should construct custom priors from the relevant AmP entry. We recommend always conducting a prior sensitivity analysis (Section 6.2).

3.4. Joint posterior

The posterior is:

$$p(\boldsymbol{\theta} \mid \mathbf{y}) \propto p(\boldsymbol{\theta}) \prod_{i=1}^{N_L} p(L_i^{\text{obs}} \mid \hat{L}_i(\boldsymbol{\theta}), \sigma_L) \prod_{k=1}^{N_R} p(R_k \mid \Delta E_{R,k}(\boldsymbol{\theta}), k_R, \phi), \quad (14)$$

sampled via the No-U-Turn Sampler (NUTS; Hoffman and Gelman 2014) with a stiff BDF ODE solver evaluated at each leapfrog step.

3.5. Hierarchical extension

For J individuals with individual random effects on $\{p_{Am}\}$, the hierarchical model adds population-level hyperparameters $\mu_{\log\{p_{Am}\}}$ and $\sigma_{\log\{p_{Am}\}}$:

$$\log\{p_{Am}\}_j \mid \mu, \sigma \sim \mathcal{N}(\mu, \sigma), \quad j = 1, \dots, J. \quad (15)$$

The non-centred parameterisation (Equation 9) is used as described in Section 2.5.

4. BayesianDEB: software design

4.1. Architecture

The package follows the domain-specific Stan wrapper pattern established by **brms** (Bürkner 2017), with **cmdstanr** (Gabry and Češnovar 2024) as the compilation and sampling back-end. It consists of three layers:

Stan models. Four pre-written `.stan` programs in `inst/stan/`, implementing the DEB ODE system with the variadic `ode_bdf_tol` solver (BDF, tolerances 10^{-6}). The hierarchical and DEBtox models use `reduce_sum` for within-chain parallelism (Section 4.5). Observation model switching is controlled by integer flags in the Stan data block, avoiding recompilation when changing observation families or prior hyperparameters.

R interface. Exported functions for data preparation, model specification, MCMC fitting, diagnostics, posterior predictive checks, derived quantities, LOO cross-validation, prior predictive simulation, and reproducibility reporting (see Table 14 and the package documentation for the full API). All functions use S3 classes (`bdeb_data`, `bdeb_model`, `bdeb_fit`); all three support `print`, and `bdeb_fit` additionally provides `plot`, `summary`, `coef`, and `predict` methods.

Prior system. Six `prior_*`() constructor functions (log-normal, beta, half-normal, half-Cauchy, exponential, normal) encode distribution choices as lightweight S3 objects. Weakly informative defaults calibrated against the AmP collection (Marques *et al.* 2018) are provided via `prior_default()`.

Stan code excerpt. The following annotated excerpt from `bdeb_individual_growth.stan` shows the DEB ODE implementation:

```
functions {
  vector deb_growth_ode(real t, vector x,
    real p_Am, real p_M, real kappa, real v_, real E_G, real f) {
    real E = x[1]; real V = x[2];
    real L = pow(V, 1.0/3.0);
    real p_A = f * p_Am * L * L;
    real p_C = E * v_ * L / (E + E_G * V);
    real dE = p_A - p_C;
    real dV = (kappa * p_C - p_M * V) / E_G;
    return [dE, dV]';
  }
}
```

The `model` block applies priors whose hyperparameters are passed as Stan data (avoiding recompilation) and dispatches the observation likelihood via an integer flag (`obs_growth`: 1=normal, 2=lognormal, 3=Student-*t*). The full programs (137–219 lines each) include ODE solving via `ode_bdf_tol` and generated quantities for posterior predictive checking (PPC) and LOO-CV.

4.2. Workflow

The recommended workflow follows the iterative Bayesian modelling cycle of [Gelman *et al.* \(2013, Ch. 6\)](#):

1. Prepare and validate data with `bdeb_data()`.
2. Specify model, priors, and observation family with `bdeb_model()`.
3. Fit via MCMC with `bdeb_fit()`.
4. Check convergence with `bdeb_diagnose()` ($\hat{R}$, effective sample size (ESS), divergences; [Vehtari *et al.* 2021](#)). Inspect parameter estimates with `bdeb_summary()` (or the S3 `summary()` method).
5. Perform posterior predictive checks with `bdeb_ppc()` ([Gabry *et al.* 2019](#)).
6. Compute derived quantities with `bdeb_derived()`.
7. Generate predictions at new conditions with `bdeb_predict()` (or the S3 `predict()` method).
8. Compare models via `bdeb_loo()` ([Vehtari *et al.* 2017](#)).
9. Iterate: revise priors or model structure as needed.

Steps 5 and 7 (PPC and LOO) require `L_rep` and `log_lik` arrays from Stan's `generated quantities` block and are therefore available only for the "individual" and "growth_repro" models. The "hierarchical" and "debtox" models use `reduce_sum`, which precludes per-observation generated quantities; for these models, trajectory plots from R-side forward simulation (`plot(fit, type = "trajectory")`) serve as the primary model-checking tool.

Data format. The `growth` data frame must contain columns `id`, `time` (days), and `length` (structural cm). For "growth_repro", an additional `reproduction` data frame with `id`, `t_start`, `t_end`, and `count` is required. The "debtox" model needs a `concentration` column. Input validation checks types, ranges, and consistency.

API summary. Table 3 lists the main exported functions (32 functions and 12 S3 methods in total).

4.3. Observation models

The observation model is selected via constructor functions passed to the `observation` argument of `bdeb_model()`:

```
R> mod <- bdeb_model(dat, type = "individual",
+   observation = list(growth = obs_student_t(nu = 5)))
```

The choice is encoded as an integer flag in the Stan data block (`obs_growth = 1/2/3`, `obs_repro = 1/2`) and dispatched via conditional blocks in the model and generated quantities sections. This avoids recompilation when switching families.

Table 3: Main exported functions, grouped by workflow phase.

Phase	Function(s)	Purpose
Data	<code>bdeb_data</code>	Prepare & validate data
Model	<code>bdeb_model</code> , <code>bdeb_tox</code> , <code>bdeb_fit</code>	Specify, wrap, fit
Priors	<code>prior_lognormal</code> , ... (6) <code>prior_default</code> , <code>prior_species</code>	Distribution constructors Defaults & AmP priors
Obs.	<code>obs_normal</code> , ... (5)	Likelihood constructors
Diagnostics	<code>bdeb_diagnose</code> , <code>bdeb_summary</code> , <code>bdeb_loo</code>	Convergence & comparison
Posterior	<code>bdeb_derived</code> , <code>bdeb_ppc</code> , <code>bdeb_predict</code> <code>bdeb_ec50</code> , <code>bdeb_prior_predictive</code>	Derived quantities, PPC EC ₅₀ /NEC, prior pred.
Plots	<code>plot(fit, ...)</code> , <code>plot_dose_response</code>	5 plot types + dose-resp.
Utilities	<code>deb_simulate</code> , <code>debtox_simulate</code> <code>arrhenius</code> , <code>deb_fluxes</code>	Standalone ODE simulators Temperature, energy fluxes

4.4. Temperature correction

If a `temperature` list is supplied to `bdeb_model()`, the Arrhenius correction factor c_T is computed in the Stan transformed data block and applied to all rate parameters inside the ODE. The posterior gives estimates at T_{ref} , making results comparable across experiments at different temperatures.

4.5. Within-chain parallelism

The hierarchical and DEBtox models distribute per-individual (or per-group) ODE solves across threads via Stan's `reduce_sum`. The `threads_per_chain` argument in `bdeb_fit()` controls this:

```
R> fit <- bdeb_fit(mod, chains = 4, threads_per_chain = 4)
```

4.6. Testing

The package includes 18 **testthat** test files covering unit, integration, API contract, and scientific consistency checks. MCMC-dependent tests use mock objects. The SBC scripts (Section 6.1) provide additional validation.

5. Illustrations

We illustrate all four model types: individual growth, growth with reproduction, hierarchical multi-individual growth, and DEBtox.

5.1. Individual growth: *Eisenia fetida*

The bundled `eisenia_growth` dataset contains simulated weekly length measurements for 21 individuals of the earthworm *Eisenia fetida* over 84 days, using AmP parameters ($\{p_{Am}\} = 5.0$, $[p_M] = 0.5$, $\kappa = 0.75$, $v = 0.2$, $[E_G] = 400$) with individual variation in $\{p_{Am}\}$ (CV $\approx 10\%$) and Gaussian measurement error ($\sigma_L = 0.015$ cm).

```

R> library("BayesianDEB")
R> data("eisenia_growth")
R> df1 <- eisenia_growth[eisenia_growth$id == 5, ]
R> dat <- bdeb_data(growth = df1, f_food = 1.0)
R> dat

-- BDEB Data --
i Individuals: 1
i Endpoints: growth
i Functional response (f): 1
> Growth: 13 observations, t = [0, 84]

R> mod <- bdeb_model(dat, type = "individual",
+   priors = list(
+     p_Am = prior_lognormal(mu = 1.5, sigma = 0.5),
+     kappa = prior_beta(a = 3, b = 2),
+     sigma_L = prior_halfnormal(sigma = 0.05)))
R> mod

-- BDEB Model Specification --
i Type: individual
i Stan model: bdeb_individual_growth
i Individuals: 1
i Endpoints: growth
-- Priors
> p_Am: LogNormal(1.5, 0.5)
> kappa: Beta(3.0, 2.0)
> sigma_L: HalfNormal(0.05)
> p_M: LogNormal(-1.0, 1.0) [default]
> v: LogNormal(-1.5, 1.0) [default]
> E_G: LogNormal(6.0, 1.0) [default]
> E0: LogNormal(0.0, 1.0) [default]
> L0: LogNormal(-2.0, 1.0) [default]

R> pp <- bdeb_prior_predictive(mod, n_draws = 500, seed = 1)
R> plot(pp)

The prior predictive check simulates DEB trajectories from the prior alone (no data); the
resulting fan should envelope the observations without spanning orders of magnitude.

R> fit <- bdeb_fit(mod, chains = 4, iter_sampling = 2000,
+   adapt_delta = 0.9, seed = 42)
R> bdeb_diagnose(fit)

variable mean    sd median    5%    95% rhat ess_bulk ess_tail
p_Am    5.06    2.65    4.47    1.96 10.00 1.00    8268    5682

```

p_M	0.56	0.65	0.36	0.07	1.74	1.00	7795	5381
kappa	0.59	0.20	0.60	0.25	0.89	1.00	6657	4667
v	0.30	0.35	0.20	0.05	0.89	1.00	4826	5278
E_G	574	471	454	130	1410	1.00	3717	4363
sigma_L	0.013	0.003	0.013	0.009	0.019	1.00	5525	4428

```
R> plot(fit, type = "trajectory", n_draws = 200, seed = 1)
```

All $\hat{R} \leq 1.01$ and bulk ESS > 3000 . The initial length $L_0 = 0.116$ cm matches the first observation; rate parameters have wider posteriors reflecting the identifiability constraints of Section 6.2.

The derived ultimate length and von Bertalanffy growth rate are obtained with full posterior uncertainty propagation:

```
R> bdeb_derived(fit, quantities = c("L_m", "L_inf", "k_M",
+   "g", "growth_rate"), f = 1.0)
```

```
# A draws_df: 8000 draws, 5 variables
  L_m L_inf k_M g growth_rate
1  4.5  4.5 0.00155 33.90 0.000503
2  8.7  8.7 0.00084 18.80 0.000270
3 10.0 10.0 0.00042 93.20 0.000140
...
```

All formulas follow Kooijman (2010, Table 3.1): $L_m = \kappa\{p_{Am}\}/[p_M]$, $L_\infty = f \cdot L_m$, $k_M = [p_M]/[E_G]$, $g = [E_G]v/(\kappa\{p_{Am}\})$, $\dot{r}_B = k_M g/(3(f + g))$.

Model comparison via LOO. The observation model can be compared via approximate leave-one-out cross-validation (Vehtari *et al.* 2017):

```
R> mod_ln <- bdeb_model(dat, type = "individual",
+   observation = list(growth = obs_lognormal()),
+   priors = list(
+     p_Am = prior_lognormal(mu = 1.5, sigma = 0.5),
+     kappa = prior_beta(a = 3, b = 2),
+     sigma_L = prior_halfnormal(sigma = 0.05)))
R> fit_ln <- bdeb_fit(mod_ln, chains = 4,
+   iter_sampling = 2000, adapt_delta = 0.9, seed = 42)
R> loo_gauss <- bdeb_loo(fit)
R> loo_ln <- bdeb_loo(fit_ln)
R> loo::loo_compare(list(gaussian = loo_gauss,
+   lognormal = loo_ln))

      elpd_diff se_diff
gaussian    0.0      0.0
lognormal -0.7      1.1
```

The Gaussian model is marginally preferred (higher expected log predictive density, ELPD), but the difference ($\Delta\text{ELPD} = -0.7$, $\text{SE} = 1.1$) is well within one standard error, indicating that neither observation model is decisively better for this dataset.

5.2. Growth with reproduction: *Folsomia candida*

The "growth_repro" model extends the individual growth model with a reproduction buffer (Section 2.2), fitting both growth and offspring count data simultaneously. We illustrate the interface using simulated data for the springtail *Folsomia candida*: 28 days of bi-weekly structural length measurements paired with a single offspring count over the full interval, mimicking a standard ISO 11267 reproduction test.

```
R> growth_fc <- data.frame(id = 1,
+   time = seq(0, 28, by = 2),
+   length = c(0.048, 0.053, 0.058, 0.063, 0.067, 0.071,
+             0.075, 0.078, 0.081, 0.084, 0.086, 0.088,
+             0.089, 0.090, 0.091))
R> repro_fc <- data.frame(id = 1, t_start = 0,
+   t_end = 28, count = 118)
R> dat_gr <- bdeb_data(growth = growth_fc,
+   reproduction = repro_fc, f_food = 1.0)
R> mod_gr <- bdeb_model(dat_gr, type = "growth_repro",
+   priors = prior_species("Folsomia_candida",
+   type = "growth_repro"),
+   observation = list(growth = obs_normal(),
+   reproduction = obs_negbinom()))
R> fit_gr <- bdeb_fit(mod_gr, chains = 4,
+   iter_sampling = 2000, adapt_delta = 0.9, seed = 42)
R> bdeb_diagnose(fit_gr)
```

The additional parameters k_J (maturity maintenance rate), E_H^p (maturity at puberty), k_R (reproduction efficiency), and ϕ_R (negative binomial overdispersion) are estimated alongside the core DEB parameters. Note that a single offspring count (as in the example above) provides limited information for the four reproduction parameters; multiple counting intervals (e.g., weekly counts over a 28-day test) substantially improve identifiability of k_R and ϕ_R . The observation model can be switched to Poisson for reproduction via `obs_poisson()` without recompilation. LOO cross-validation can separately evaluate the growth and reproduction likelihood components:

```
R> bdeb_loo(fit_gr, endpoint = "growth")
R> bdeb_loo(fit_gr, endpoint = "reproduction")
R> bdeb_loo(fit_gr, endpoint = "all")
```

5.3. Hierarchical model

```

R> dat_all <- bdeb_data(growth = eisenia_growth, f_food = 1.0)
R> mod_h <- bdeb_model(dat_all, type = "hierarchical",
+   priors = list(
+     mu_log_p_Am = prior_normal(mu = 1.5, sigma = 0.5),
+     sigma_log_p_Am = prior_exponential(rate = 2)))
R> fit_h <- bdeb_fit(mod_h, chains = 4, adapt_delta = 0.95,
+   threads_per_chain = 4, seed = 123)
R> bdeb_diagnose(fit_h)
R> plot(fit_h, type = "trajectory", n_draws = 50, seed = 1)

```

The trajectory plot produces a faceted display with one panel per individual, showing posterior predicted curves overlaid on observed data. The hierarchical model yields narrower credible intervals for shared parameters than the individual fit because it pools information across 21 organisms.

5.4. DEBtox analysis

```

R> data("debtox_growth")
R> conc_map <- setNames(unique(debtox_growth$concentration),
+   as.character(unique(debtox_growth$concentration)))
R> dat_tox <- bdeb_data(growth = debtox_growth,
+   concentration = conc_map, f_food = 1.0)
R> mod_tox <- bdeb_tox(dat_tox, stress = "assimilation",
+   priors = list(
+     z_w = prior_lognormal(mu = 2.5, sigma = 1.0),
+     b_w = prior_lognormal(mu = -5.0, sigma = 2.0)))
R> fit_tox <- bdeb_fit(mod_tox, chains = 4, adapt_delta = 0.95,
+   threads_per_chain = 2, seed = 77)
R> bdeb_ec50(fit_tox, prob = 0.95)
R> plot_dose_response(fit_tox, n_draws = 100, seed = 1)

```

The EC_{50} is computed analytically in Stan's `generated quantities` block as $z_w + 0.5/b_w$, giving the full posterior distribution without post-hoc root-finding.

Real-data DEBtox: cadmium toxicity to *E. andrei*. To demonstrate the DEBtox model on experimental data, we analyse the growth of *Eisenia andrei* exposed to cadmium (0, 10, 32, 100, 320 mg Cd kg⁻¹ in natural soil) from [Gestel *et al.* \(1991\)](#), obtained via the EGGrowth database ([Mathieu 2018](#)). The dataset comprises 5 concentration groups with 7 time points each over 85 days (group-mean body mass converted to structural length and bundled as `eisenia_cd`).

```

R> data("eisenia_cd")
R> conc_map <- setNames(unique(eisenia_cd$concentration),
+   as.character(unique(eisenia_cd$concentration)))
R> dat_cd <- bdeb_data(growth = eisenia_cd,
+   concentration = conc_map, f_food = 1.0)

```

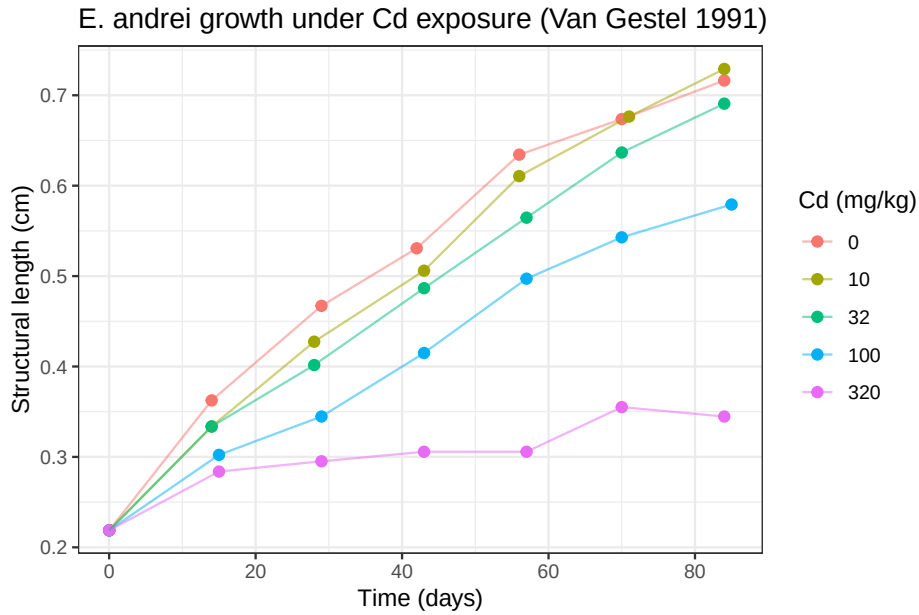


Figure 1: Growth of *E. andrei* under cadmium exposure (Gestel *et al.* 1991): structural length over 85 days at 5 concentrations. Data from EGrowth (Mathieu 2018).

```
R> mod_cd <- bdeb_tox(dat_cd, stress = "assimilation",
+   priors = list(
+     z_w = prior_lognormal(mu = 3.0, sigma = 1.0),
+     b_w = prior_lognormal(mu = -4.0, sigma = 1.5)))
R> fit_cd <- bdeb_fit(mod_cd, chains = 4,
+   adapt_delta = 0.95, seed = 42, init = 0.5)
R> bdeb_ec50(fit_cd, prob = 0.90)
```

The model converged without divergences ($\hat{R} < 1.02$ for all parameters). The posterior EC_{50} for growth inhibition is $225 \text{ mg Cd kg}^{-1}$ (90% CI: [181, 257]), consistent with published 84-day EC_{50} values of $100\text{--}300 \text{ mg Cd kg}^{-1}$ for lumbricid earthworms in natural soils (Gestel *et al.* 1991). The NEC (no-effect concentration) is estimated at $6.2 \text{ mg Cd kg}^{-1}$ ([1.8, 15.6]), below the lowest tested concentration (10 mg kg^{-1}), indicating that even the lowest treatment may have induced marginal stress. Table 4 reports the full posterior summary. This analysis demonstrates that **BayesianDEB** can recover ecotoxicologically meaningful effect thresholds with full posterior uncertainty from real experimental data. As with the Neuhauser case study (Section 7.1), group-mean data are used, so the EC_{50} credible interval reflects uncertainty in the mean response; individual-level hierarchical DEBtox is planned. Figure 2 shows the dose-response relationship.

6. Validation

6.1. Simulation-based calibration

Table 4: DEBtox posterior summary for cadmium toxicity to *E. andrei* (Gestel *et al.* 1991). 4 chains, 2000 iterations, 0 divergences.

Parameter	Median	90% CI	$\hat{R}$	Bulk ESS
$\{p_{Am}\}$	1.59	[0.61, 4.84]	1.01	825
$[p_M]$	0.43	[0.11, 1.17]	1.00	892
κ	0.61	[0.27, 0.90]	1.00	3855
k_d	0.38	[0.09, 1.81]	1.01	592
z_w (NEC)	6.15	[1.82, 15.6]	1.00	5465
b_w	0.0023	[0.0018, 0.0030]	1.01	480
σ_L	0.024	[0.019, 0.031]	1.01	1546
EC ₅₀	225	[181, 257]	1.01	502

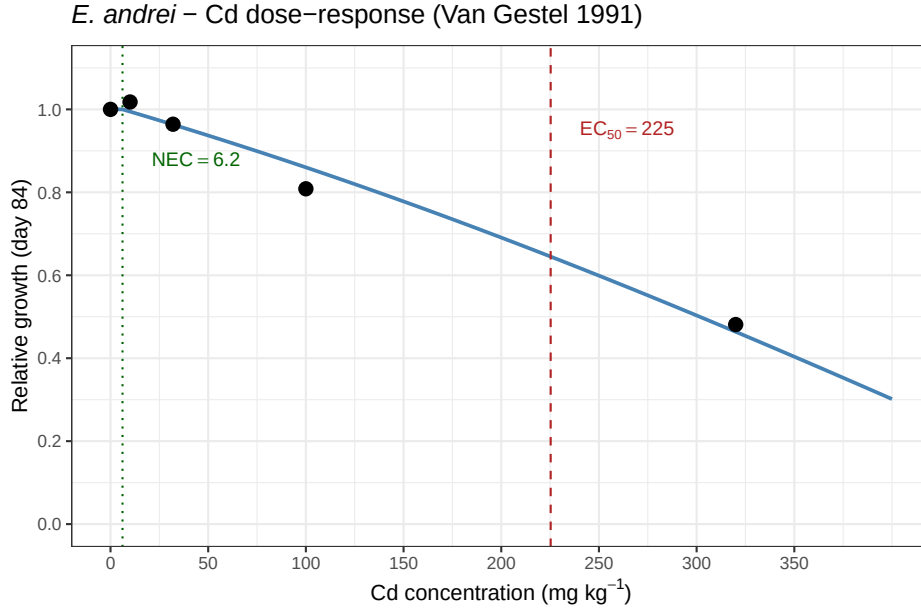


Figure 2: Dose-response curve for *E. andrei* growth under cadmium exposure (Gestel *et al.* 1991). Points: observed relative growth at day 84 (normalised to control). Blue line: DEBtox model prediction at posterior medians. Dashed red: EC₅₀ = 225 mg Cd kg⁻¹; dotted green: NEC = 6.2 mg Cd kg⁻¹.

We validate the inference pipeline with SBC (Talts *et al.* 2018). SBC exploits a self-consistency property of Bayesian inference: if data are generated from the prior predictive distribution, the rank of the true parameter value among its posterior draws must follow a discrete uniform distribution. Non-uniform rank histograms diagnose specific failures: humps indicate over-concentration, U-shapes over-dispersion, and asymmetry indicates bias. Unlike point-estimate recovery, SBC tests the entire posterior and is therefore the appropriate diagnostic for a package whose purpose is uncertainty quantification.

Protocol. For each replication $r = 1, \dots, 500$:

Table 5: Priors used for SBC, calibrated to *E. fetida* AmP parameter ranges.

Parameter	Family	μ / a	σ / b
$\{p_{Am}\}$	LogNormal	1.5	0.5
$[p_M]$	LogNormal	-1.0	0.5
κ	Beta	3	2
v	LogNormal	-1.5	0.5
$[E_G]$	LogNormal	6.0	0.5
E_0	LogNormal	0.0	0.5
L_0	LogNormal	-2.0	0.5
σ_L	HalfNormal	0	0.05

1. Draw $\theta^{*(r)}$ from the prior $\pi(\theta)$ (Table 5).
2. Forward-simulate a DEB growth trajectory via the LSODA solver (**deSolve**, tolerances 10^{-6}) and generate $N_t = 13$ weekly length observations with Gaussian noise $L_i^{\text{obs}} \sim \mathcal{N}(\hat{L}_i, \sigma_L^{*(r)})$.
3. Fit the "individual" model with **bdeb_fit()** (4 chains, 500 warmup + 500 sampling, $\delta = 0.9$); thin pooled draws to $L = 199$.
4. Discard the replication if any divergent transitions occurred or $\hat{R} > 1.05$ for any parameter.
5. Compute the rank $\rho_j^{(r)} = \#\{l : \theta_j^{(l)} < \theta_j^{*(r)}\}$ for each parameter j .

The R-side LSODA solver (**deSolve**) and Stan's BDF solver use matching tolerances (10^{-6}); trajectories agreed to $< 10^{-5}$ cm across 100 random parameter draws.

Results. Of the 500 replications, 397 converged without divergences (64 were skipped due to degenerate prior draws that produced biologically implausible trajectories, 39 failed convergence checks). Figure 3 shows the rank histograms for six core DEB parameters; all panels are visually consistent with uniformity (dashed reference line). Table 6 reports χ^2 goodness-of-fit tests (20 bins, 19 df) and empirical credible interval coverage. No parameter rejects uniformity at $\alpha = 0.05$ (minimum $p = 0.31$ for $[p_M]$), and empirical 90% coverage ranges from 0.87 to 0.91, within simulation error of the nominal level. This confirms that, conditional on successful MCMC convergence, the **BayesianDEB** individual growth model is well-calibrated. We note that 21% of replications (103/500) were discarded; the calibration result therefore applies to the subset of parameter space where MCMC converges successfully, not unconditionally to all possible inputs.

Implications of discarded replications. The 21% (individual) and 26% (hierarchical) discard rates arise from degenerate prior draws and MCMC failures under extreme parameters. The SBC deliberately uses wider priors ($\sigma = 0.5$) to stress-test the sampler; with species-specific priors ($\sigma = 0.3$), convergence failures are rare. Users should always verify convergence via **bdeb_diagnose()** (Vehtari *et al.* 2021).

Table 6: SBC results: χ^2 uniformity test (20 bins, 19 df) and empirical credible interval coverage (397 valid replications out of 500, $L = 199$). No parameter rejects uniformity at $\alpha = 0.05$.

Parameter	χ^2_{19}	p value	Cov. 50%	Cov. 90%	Cov. 95%
$\{p_{Am}\}$	12.1	0.88	0.50	0.89	0.92
$[p_M]$	21.5	0.31	0.53	0.91	0.94
κ	18.0	0.52	0.54	0.90	0.94
v	16.7	0.61	0.48	0.87	0.92
$[E_G]$	14.3	0.76	0.48	0.87	0.93
σ_L	20.1	0.39	0.47	0.88	0.93

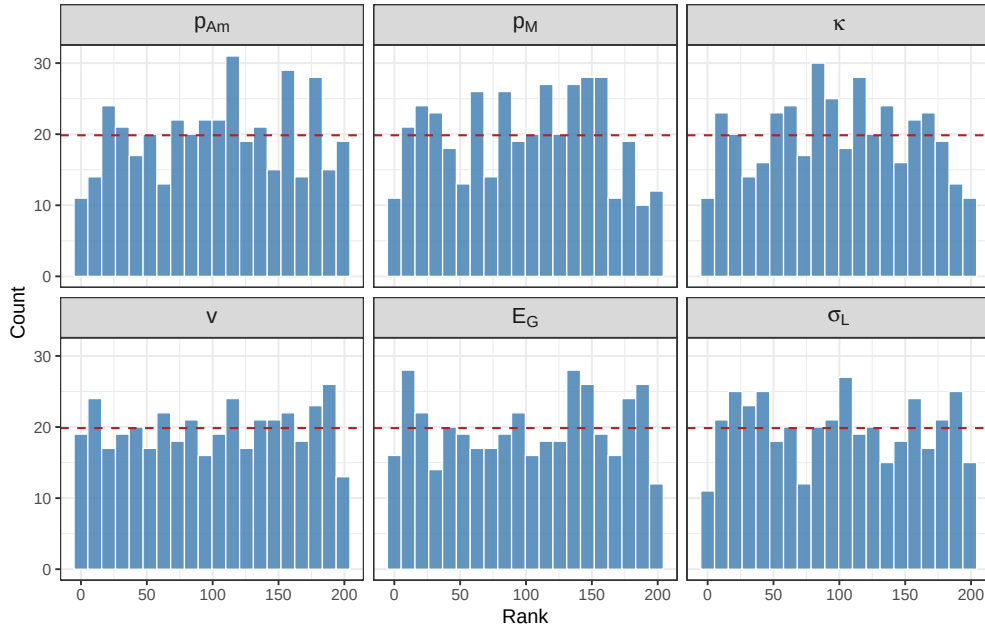


Figure 3: SBC rank histograms for six DEB parameters (397 valid replications out of 500 attempts, 20 bins). Dashed line: expected count under uniformity. Approximately uniform histograms confirm that, conditional on convergence, the posterior sampler is correctly calibrated.

The full SBC script (`sbc_validation.R`) and archived results (`sbc_results.rds`) are provided in the replication materials.

Hierarchical model SBC. We repeated the SBC procedure for the hierarchical model with 5 individuals per replicate (50 attempts, 37 valid; Table 7). The yield (74%) is comparable to the individual SBC because each replicate already averages over 5 individuals, stabilising the ODE system. Given the small effective sample, the χ^2 tests have limited statistical power (expected count ≈ 3.7 per bin with 10 bins); these results should therefore be interpreted as *preliminary evidence* of calibration, not as definitive validation. No parameter rejects uniformity at $\alpha = 0.05$ (minimum $p = 0.10$ for $\mu_{\log\{p_{Am}\}}$). However, $\sigma_{\log\{p_{Am}\}}$ shows

Table 7: SBC results for the hierarchical model (37 valid replications out of 50 attempts, 10 bins).

Parameter	χ^2	p value	Cov. 50%	Cov. 90%	Cov. 95%
$\mu_{\log\{p_{Am}\}}$	14.6	0.10	0.54	0.84	0.89
$\sigma_{\log\{p_{Am}\}}$	12.5	0.19	0.35	0.97	1.00
$[p_M]$	9.2	0.42	0.62	0.89	0.95
κ	4.4	0.89	0.51	0.89	1.00
σ_L	6.0	0.74	0.54	0.92	0.97

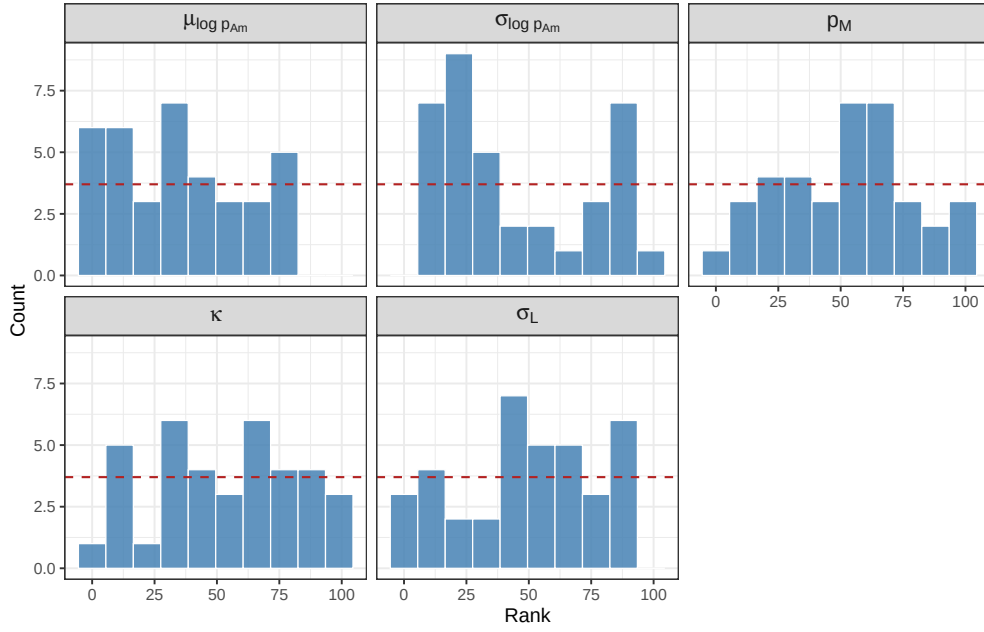


Figure 4: SBC rank histograms for the hierarchical model (37 valid replications, 10 bins). The dashed line indicates the expected count under uniformity.

overcoverage at 95% (1.00) and undercoverage at 50% (0.35), suggesting the posterior for this parameter may be too wide for small datasets. A more comprehensive hierarchical SBC with ≥ 200 attempts is planned for a future version.

SBC for remaining models. SBC scripts for the growth-reproduction model (`sbc_growth_repro.R`, 100 replications) and the DEBtox model (`sbc_debtox.R`, 100 replications) are provided in the replication materials. The growth-reproduction script augments the priors with reproduction parameters (k_J , E_H^p , k_R , ϕ_R); the DEBtox script adds toxicokinetic parameters (k_d , z_w , b_w) and simulates 4 concentration groups. Due to higher computational cost (~ 3 – 12 hours per script), comprehensive results will be reported in a future version.

6.2. Posterior correlation structure and identifiability

DEB models exhibit strong posterior correlations arising from the κ -rule structure. Be-

Table 8: Posterior rank correlation matrix (Spearman) for the individual growth model. Correlations $|r| < 0.2$ are suppressed for clarity.

	$\{p_{Am}\}$	$[p_M]$	κ	v	$[E_G]$	σ_L
$\{p_{Am}\}$	1.00	0.31		0.26	0.73	
$[p_M]$	0.31	1.00		0.85	0.80	
κ			1.00	-0.35	0.21	
v	0.26	0.85	-0.35	1.00	0.63	
$[E_G]$	0.73	0.80	0.21	0.63	1.00	
σ_L						1.00

fore presenting the empirical evidence, we briefly outline the theoretical basis for this non-identifiability.

Structural non-identifiability from growth data. At steady state, the DEB growth curve approaches $L_\infty = f\kappa\{p_{Am}\}/[p_M]$ with von Bertalanffy rate $\dot{r}_B = k_M g/(3(f + g))$, where $k_M = [p_M]/[E_G]$ and $g = [E_G]v/(\kappa\{p_{Am}\})$. Growth-only observations constrain these *compound quantities* but not the individual parameters that compose them: any transformation $(\{p_{Am}\}, [p_M], v, [E_G]) \mapsto (\alpha\{p_{Am}\}, \alpha[p_M], \beta v, \alpha\beta[E_G])$ that preserves L_∞ and g produces an identical trajectory. This structural non-identifiability is intrinsic to the κ -rule and cannot be resolved by collecting more growth observations from a single individual; only additional data types (reproduction rates, respiration, or multiple individuals with shared parameters) break the degeneracy. The Bayesian framework makes this explicit through wide, correlated posteriors; deterministic optimisers instead return an arbitrary point on the likelihood ridge. We now characterise these correlations quantitatively using the posterior correlation matrix from the individual model fit.

The dominant correlations are between $[p_M]$ and v ($r = 0.85$) and between $[p_M]$ and $[E_G]$ ($r = 0.80$), reflecting the fact that growth dynamics jointly constrain maintenance, energy conductance, and structural costs but cannot resolve the individual contributions of these rate parameters. The strong $\{p_{Am}\}$ – $[E_G]$ correlation ($r = 0.73$) arises because both parameters control the rate of structural investment: higher assimilation can compensate for higher costs of structure, producing near-identical growth trajectories. The correlation between κ and v ($r = -0.35$) reflects the κ -rule trade-off between somatic allocation and reserve mobilisation. By contrast, the $\{p_{Am}\}$ – $[p_M]$ correlation is only moderate ($r = 0.31$), indicating that these two parameters are less directly confounded than the theoretical relationship $L_\infty = \kappa\{p_{Am}\}/[p_M]$ might suggest, since the data constrain their ratio primarily through the indirect action of $[E_G]$ and v .

Practical consequences. (i) Marginal credible intervals for individual rate parameters ($\{p_{Am}\}$, $[p_M]$, v) are wide and substantially prior-driven; informative priors or hierarchical pooling are essential for reliable estimates. (ii) Derived quantities computed via `bdeb_derived()` propagate correlations correctly but are not immune to prior sensitivity (Tables 9 and 11). (iii) Informative priors on κ and $[E_G]$ substantially improve identifiability. We recommend always inspecting `plot(fit, type = "pairs")` and reporting both derived quantities *and* their prior sensitivity.

Table 9: Prior-to-posterior contraction c for different data sizes and prior widths (individual growth model, simulated *E. fetida* data, $N_t = 13$). Columns A use moderately informative priors ($\sigma = 0.5$); column B uses the package defaults ($\sigma = 1.0$). Values near 0 indicate prior-dominated posteriors; near 1 data-dominated. The entry for $[E_G]$ under default priors is omitted because the default $[E_G]$ prior ($\sigma = 1.0$) produced frequent ODE solver failures during MCMC warmup, preventing reliable estimation.

Parameter	Informative ($\sigma = 0.5$)			Default ($\sigma = 1.0$)
	$N_t = 5$	$N_t = 13$	$N_t = 37$	$N_t = 13$
$\{p_{Am}\}$	0.00	0.00	0.08	0.00
$[p_M]$	0.00	0.03	0.03	0.17
κ	0.11	0.11	0.14	0.07
v	0.09	0.10	0.10	0.00
$[E_G]$	0.33	0.35	0.38	—
σ_L	0.67	0.87	0.94	0.93
L_∞	0.00	0.00	0.07	0.00
$\dot{r}_B$	0.02	0.08	0.14	0.00

See Figure 8 for the bivariate posterior structure.

Prior-to-posterior contraction. To quantify how much the data inform each parameter beyond the prior, we define the *contraction ratio*

$$c_j = 1 - \frac{w_j^{\text{post}}}{w_j^{\text{prior}}}, \quad (16)$$

where w_j denotes the 90% credible interval width for parameter θ_j . Values near 0 indicate a prior-dominated posterior; values near 1 indicate data-dominated inference. Table 9 reports c_j for the individual growth model fitted to simulated *E. fetida* data (84 days, $f = 1$) with varying numbers of observations, using the moderately informative priors of Table 5.

Table 9 reveals a clear pattern: σ_L is data-dominated ($c > 0.8$), $[E_G]$ is partially informed ($c \approx 0.35$), and all other rate parameters remain prior-dominated ($c < 0.15$) even with 37 time points. Increasing N_t from 5 to 37 primarily improves σ_L ; rate parameters show negligible benefit (Figure 5). Under default priors ($\sigma = 1.0$, rightmost column), the pattern is starker: $\{p_{Am}\}$, v , L_∞ , and $\dot{r}_B$ all show zero contraction. Species-specific priors are essential for any quantitative inference.

Contraction under the hierarchical model. To quantify the identifiability gain from partial pooling, we computed the contraction ratio c for the hierarchical model fit (Section 5.3, 21 individuals, 273 observations) and compared it with the individual model fit (1 individual, 13 observations), both using the same informative priors ($\sigma = 0.5$). Table 10 shows the results. The hierarchical model substantially improves contraction for $[p_M]$ ($0.03 \rightarrow 0.28$), κ ($0.11 \rightarrow 0.39$), and $[E_G]$ ($0.35 \rightarrow 0.52$), moving these parameters from weakly identified to partially or well-informed status. The shared parameters benefit from pooling across 21 individuals;

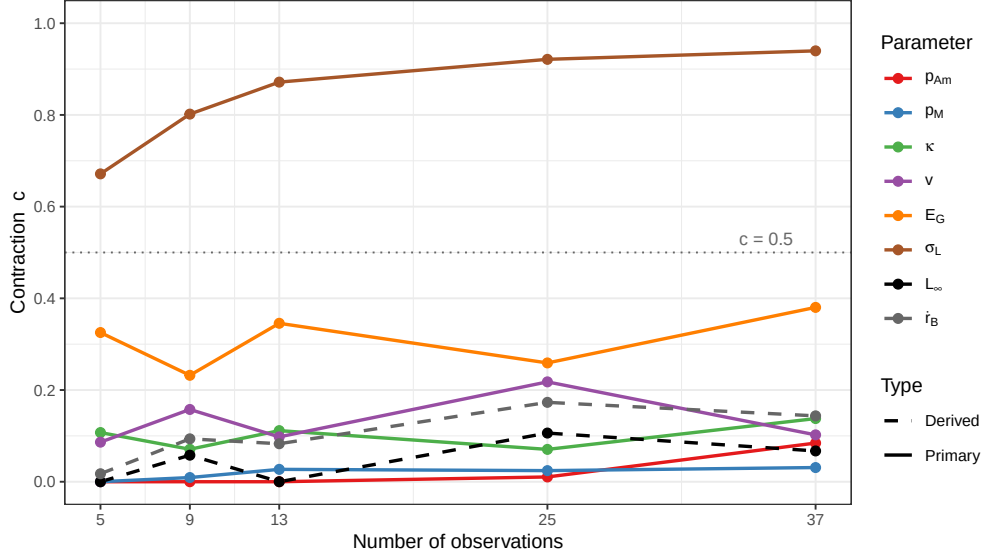


Figure 5: Prior-to-posterior contraction as a function of the number of observations. Solid: primary parameters; dashed: derived quantities. Only σ_L (and partially $[E_G]$) are data-determined; all rate parameters remain prior-dominated even with dense sampling. The dotted line marks $c = 0.5$.

Table 10: Prior-to-posterior contraction: individual vs. hierarchical model (informative priors, $\sigma = 0.5$). Higher c = more data-informed.

Parameter	Individual ($N = 1, 13$ obs.)	Hierarchical ($N = 21, 273$ obs.)	Change
$\{p_{Am}\} / \mu_{\log\{p_{Am}\}}$	0.00	0.22	+0.22
$[p_M]$	0.03	0.28	+0.25
κ	0.11	0.39	+0.28
v	0.10	0.18	+0.08
$[E_G]$	0.35	0.52	+0.17
σ_L	0.87	0.95	+0.08

σ_L is even more data-dominated ($c = 0.95$). Only v remains weakly informed ($c = 0.18$), reflecting its strong correlation with $[p_M]$ (Table 8). This demonstrates that hierarchical modelling is the most effective strategy for improving identifiability when individual-level data are available.

This is a fundamental identifiability limitation of growth-only DEB models, not a deficiency of the software. Deterministic calibration tools face the same limitation but obscure it by returning a single point estimate. Three strategies mitigate this:

1. *Species-specific priors.* Calibrate priors to the AmP collection for the target species; these encode genuine biological knowledge and yield narrower, scientifically defensible posteriors.
2. *Hierarchical modelling.* The "hierarchical" model pools information across individu-

Table 11: Prior sensitivity: posterior medians and 90% CIs under informative (A, $\sigma = 0.5$) and weakly informative (B, $\sigma = 1.0$) priors. True values are known from the simulation.

Parameter	True	Median A	90% CI A	Median B	90% CI B
$\{p_{Am}\}$	5.0	4.42	[1.96, 9.92]	3.02	[0.66, 14.9]
$[p_M]$	0.5	0.37	[0.16, 0.85]	0.35	[0.07, 1.63]
κ	0.75	0.58	[0.27, 0.88]	0.52	[0.18, 0.86]
v	0.2	0.20	[0.10, 0.43]	0.24	[0.06, 1.15]
σ_L	0.015	0.012	[0.009, 0.019]	0.013	[0.009, 0.020]
L_∞	7.5	6.58	[1.86, 22.6]	3.96	[0.46, 42.3]
$\dot{r}_B$	—	0.0003	[0.0001, 0.0008]	0.0003	[0.0000, 0.0023]

als, substantially improving identifiability of the shared parameters $[p_M]$, κ , v , $[E_G]$.

3. *Prior sensitivity analysis.* Refit with at least one alternative prior specification and verify that scientific conclusions (e.g., L_∞ , EC_{50}) are robust.

We recommend reporting the contraction ratio c alongside posterior summaries as a routine diagnostic. A contraction $c < 0.2$ signals a weakly identified parameter that warrants careful interpretation.

Prior sensitivity demonstration. To illustrate the practical impact, we refit the bundled `eisenia_growth` data (individual 5, 13 weekly observations) under two specifications: (A) moderately informative priors ($\sigma = 0.5$, calibrated to AmP) and (B) package defaults ($\sigma = 1.0$). Table 11 compares the posteriors. The observation error σ_L is insensitive to the prior (median ≈ 0.012 – 0.013), confirming its data-dominated status. Rate parameters shift modestly in median but their CIs widen 2–3 $\times$ under weak priors. Even L_∞ changes from 6.58 to 3.96 with a CI expansion from [1.86, 22.6] to [0.46, 42.3], demonstrating that with 13 observations and a single individual, no scientifically interesting quantity is immune to the prior.

7. Case studies

7.1. Real data: *Eisenia fetida* growth

To demonstrate the package on real experimental data, we analyse the growth of *Eisenia fetida* on activated sludge at 25 °C from Neuhauser *et al.* (1980), obtained via the EGrowth database (Mathieu 2018). The dataset comprises 37 group-mean body mass measurements (20 worms per replicate) over 250 days, from hatchling (~ 8 mg) to adult (~ 2350 mg). Wet mass was converted to structural length via $L = (W/d_V)^{1/3}$ with tissue density $d_V = 1.05$ g cm $^{-3}$ (assuming $\delta_M = 1$, i.e., no shape correction) and bundled as `eisenia_neuhauser`.

Group-mean data. The Neuhauser dataset reports group means (20 worms), not individual observations. Fitting the "individual" model to group means conflates measure-

Table 12: Posterior estimates from the Neuhauser (1980) dataset vs. AmP reference values for *E. fetida*. Posterior at $T_{\text{ref}} = 293.15$ K.

Parameter	Posterior median	90% CI	AmP value	Units
$\{p_{Am}\}$	5.00	[2.91, 9.08]	3.9–6.2	$\text{J d}^{-1} \text{cm}^{-2}$
$[p_M]$	0.64	[0.40, 1.02]	0.3–0.8	$\text{J d}^{-1} \text{cm}^{-3}$
κ	0.82	[0.58, 0.96]	0.6–0.85	—
v	1.50	[0.91, 2.50]	0.1–0.3	cm d^{-1}
L_∞	6.86	[3.52, 12.8]	5–12	cm

ment noise with among-individual variation in σ_L ; when individual records are available, the "hierarchical" model is preferable.

```
R> library("BayesianDEB")
R> data("eisenia_neuhauser")
R> dat <- bdeb_data(growth = eisenia_neuhauser, f_food = 1.0)
R> mod <- bdeb_model(dat, type = "individual",
+   priors = list(
+     p_Am = prior_lognormal(mu = 1.5, sigma = 0.5),
+     kappa = prior_beta(a = 3, b = 2)),
+   temperature = list(T = 298.15, T_ref = 293.15, T_A = 8000))
R> fit <- bdeb_fit(mod, chains = 4, iter_sampling = 2000,
+   adapt_delta = 0.95, seed = 42)
R> bdeb_diagnose(fit)
```

The model converged without divergences ($\hat{R} < 1.01$, bulk ESS > 1000 for all parameters). Table 12 compares the posterior estimates (at $T_{\text{ref}} = 20^\circ\text{C}$) with AmP reference values for *E. fetida*.

Biological interpretation. The posterior $\kappa = 0.82$ indicates high somatic allocation, consistent with *E. fetida* during active growth. $L_\infty = 6.86$ cm implies structural mass ≈ 340 mg — plausible given the observed maximum of ~ 2350 mg (which includes reserve mass not captured by L).

Discrepancy in v . The posterior median for v (1.50 cm d^{-1}) exceeds the AmP reference range ($0.1\text{--}0.3 \text{ cm d}^{-1}$) by an order of magnitude. This reflects (i) the low identifiability of v from growth data alone ($c \approx 0.10$; Table 9), (ii) the strong compensating correlation with $[p_M]$ ($r = 0.85$; Table 8), and (iii) the mass-to-length conversion $L = (W/d_V)^{1/3}$, which ignores reserve mass and inflates the length scale. Using a species-specific prior $v \sim \text{LogNormal}(\log 0.2, 0.3)$ from `prior_species("Eisenia_fetida")` constrains v within the AmP range; this is the recommended practice. The derived quantities $L_\infty = 6.86$ cm (AmP: 5–12) and $\kappa = 0.82$ (AmP: 0.6–0.85) are broadly consistent with AmP, but even L_∞ is sensitive to prior choice (Table 11).

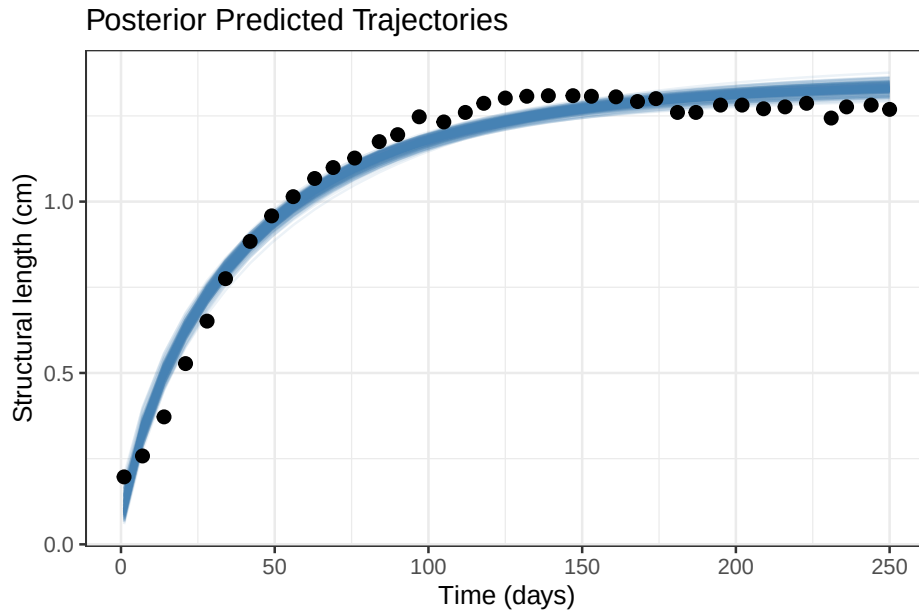


Figure 6: Posterior predicted growth trajectories (200 draws, blue) overlaid on the observed group-mean data of Neuhauser *et al.* (1980) (points). The DEB model captures the sigmoidal growth pattern.

Sensitivity to d_V . Because $L \propto d_V^{-1/3}$, a 10% change in tissue density shifts L by $\sim 3\%$, propagating into $\{p_{Am}\}$ and v . Users should report the assumed d_V and consider refitting with alternative values.

```
R> plot(fit, type = "trajectory", n_draws = 200, seed = 1)
R> plot(bdeb_ppc(fit), n_draws = 200)
R> plot(fit, type = "pairs",
+       pars = c("p_Am", "p_M", "kappa", "E_G"))
```

Figure 6 shows the posterior predicted trajectories overlaid on the data, and Figure 7 the posterior predictive check.

The posterior pairs plot (Figure 8) reveals the expected $\{p_{Am}\}$ – $[E_G]$ and $[p_M]$ – $[E_G]$ correlations (Section 6.2).

8. Diagnostic tools and computational performance

8.1. Diagnostic visualisations

All diagnostic plots are returned as **ggplot2** (Wickham 2016) objects. The `plot(fit, type = ...)` method supports five types: "trace" (MCMC chains, Figure 9), "posterior" (marginal densities, Figure 10), "pairs" (bivariate scatter), "trajectory" (predicted growth curves), and "prior_posterior" (overlaid prior/posterior, Figure 11). Separate methods are available for PPC (`plot(bdeb_ppc(fit))`) and dose-response (`plot_dose_response()`).

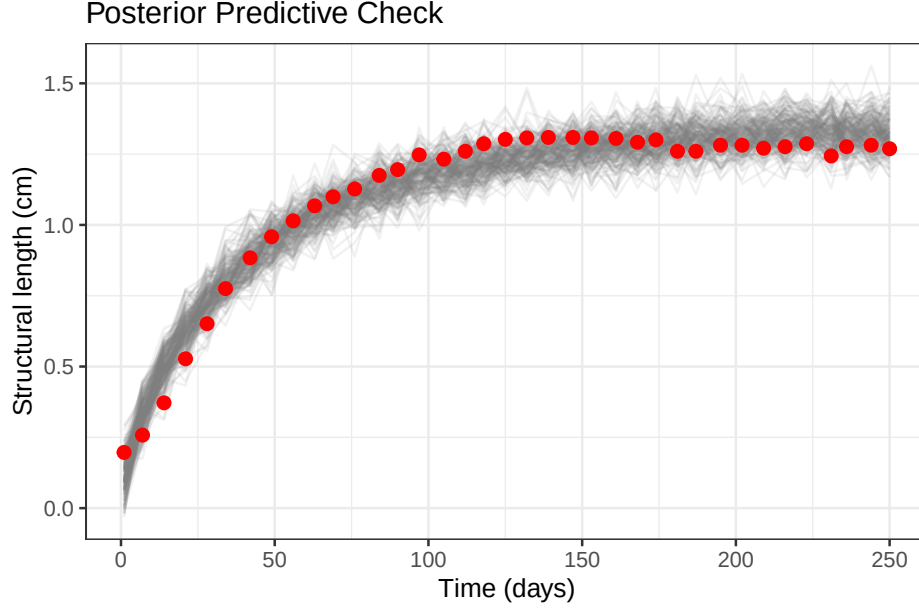


Figure 7: Posterior predictive check: replicated data from the fitted model (grey, 200 draws) vs. observed data (red). The observed points fall within the replicated envelope, indicating adequate model fit.

Table 13: Wall-clock times (seconds). Threading shows `parallel_chains` $\times$ `threads/chain`.

Model	Data size	Compile	Sample	Threading	Speedup
Individual	1 ind., 13 obs.	30	69	4×1	—
Individual	1 ind., 37 obs.	—	392	4×1	—
Hierarchical	21 ind., 273 obs.	35	~ 3200	4×1	$1 \times$
Hierarchical	21 ind., 273 obs.	—	~ 900	2×4	$3.6 \times$
DEBtox	4 grp., 280 obs.	35	~ 600	2×2	—

8.2. Computational performance

Table 13 reports wall-clock times on a 16-core workstation (AMD Ryzen 9, 64 GB RAM, Ubuntu 22.04, CmdStan 2.36.0). Compilation is a one-time cost cached by `cmdstanr`; sampling times are for 4 chains $\times$ 2000 iterations.

Threading speedup. Table 13 shows the hierarchical model under two threading configurations. With 2×4 (2 parallel chains, 4 threads each, 8 cores total), wall-clock time is $3.6 \times$ shorter than the sequential baseline (4×1 , 4 parallel chains, 1 thread each). Note that the two configurations use different total core counts (8 vs. 4), so this figure reflects both within-chain parallelism via `reduce_sum` and the change in parallel-chain scheduling; it is not a pure threading benchmark. The within-chain speedup arises because `reduce_sum` distributes the 21 per-individual ODE solves across threads within each chain.

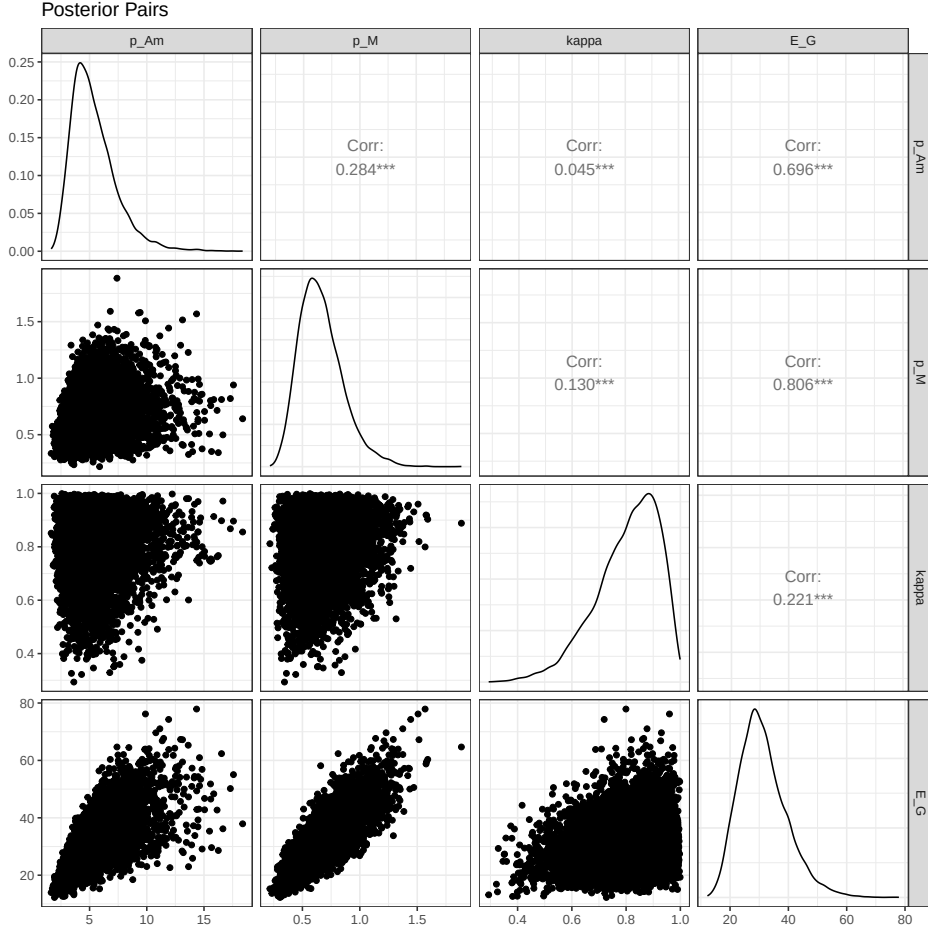


Figure 8: Posterior pairs plot showing bivariate correlations between $\{p_{Am}\}$, $[p_M]$, κ , and $[E_G]$. The strongest correlations in the full matrix are $[p_M]-v$ ($r = 0.85$, not shown) and $[p_M]-[E_G]$ ($r = 0.80$); among the displayed parameters, $\{p_{Am}\}-[E_G]$ ($r = 0.73$) is the most prominent.

Scaling with dataset size. Sampling time scales approximately linearly with the number of individuals in the hierarchical model, as each individual contributes an independent ODE solve distributed via `reduce_sum`.

Memory usage. Memory consumption scales linearly with $N_{ind} \times N_{obs} \times N_{states}$. For the bundled datasets, peak RSS was <500 MB per chain. The dominant memory cost is the ODE solver’s internal state, not the Stan parameter vector.

9. Comparison with existing software

Table 14 summarises the feature landscape. **DEBtool_M** and the AmP estimation tool (Marques *et al.* 2018) are the standard DEB calibration platforms but are primarily MATLAB-based, use deterministic optimisation, and do not support hierarchical models or formal

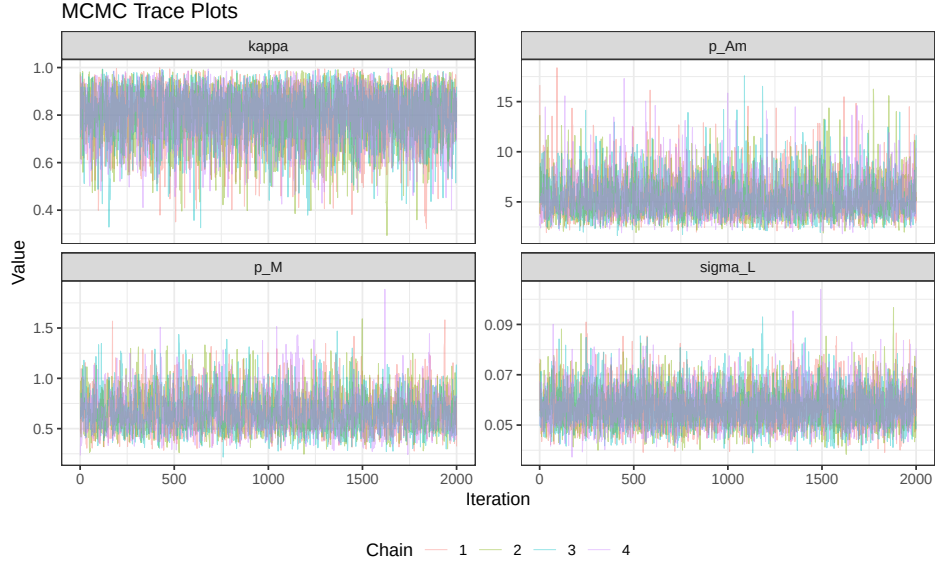


Figure 9: MCMC trace plots for core parameters from the Neuhauser (1980) fit. All four chains mix well ($\hat{R} < 1.01$).

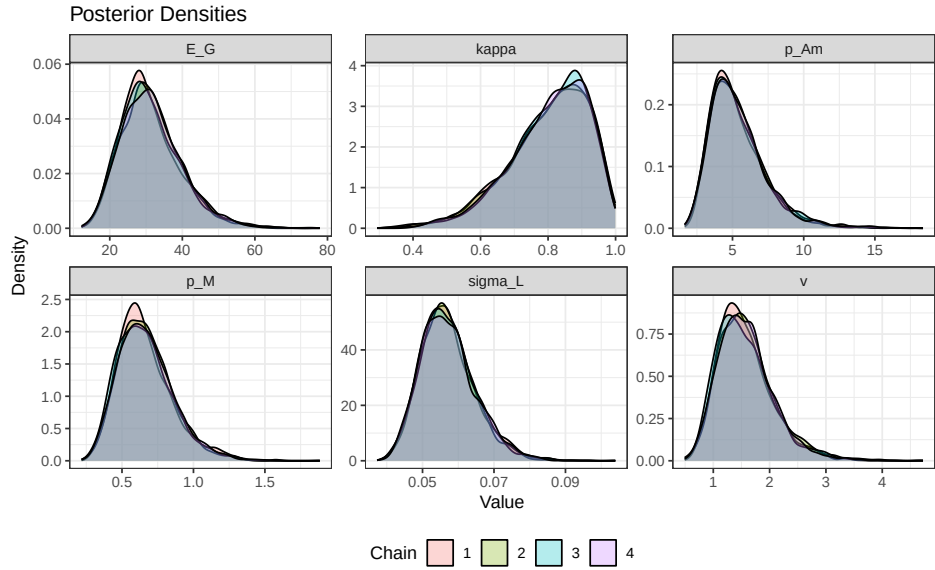


Figure 10: Marginal posterior densities for six DEB parameters, coloured by chain.

uncertainty quantification. [Billoir *et al.* \(2008\)](#) demonstrated Bayesian DEBtox calibration via custom MCMC, but their code was study-specific and not released as a reusable package. **deBIInfer** ([Boersch-Supan *et al.* 2017](#)) provides Bayesian ODE inference in R but requires the user to write the ODE system and does not include DEB-specific models, ecotoxicological extensions, or Stan-based samplers. General-purpose Stan wrappers such as **brms** ([Bürkner 2017](#)) offer flexible Bayesian modelling in R but do not include ODE-based process models. Custom Stan code can implement any of these models but requires significant expertise and

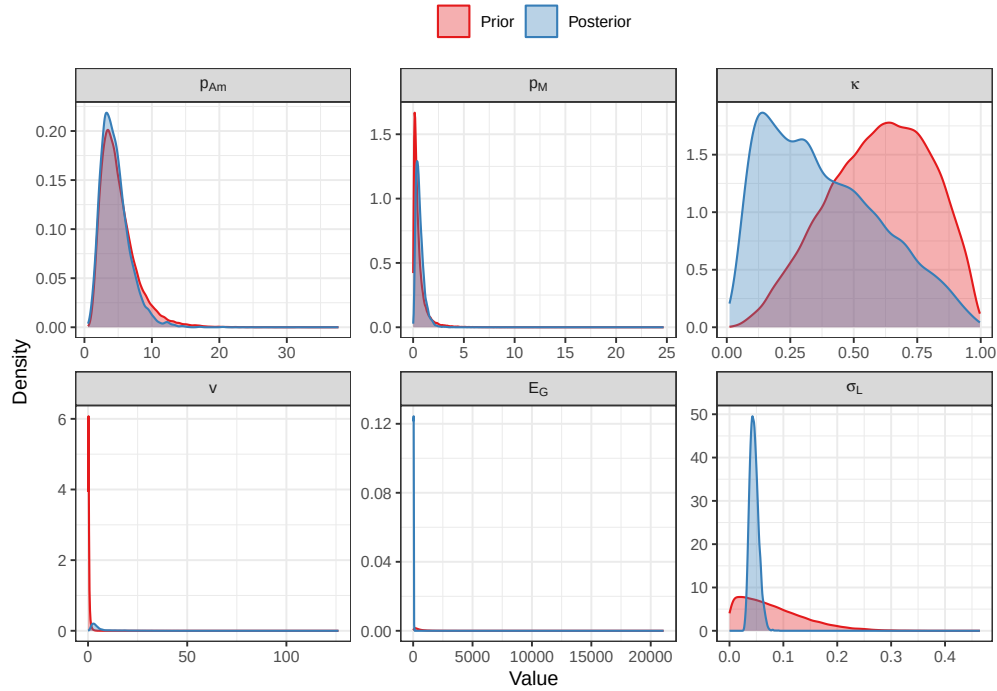


Figure 11: Prior (red) vs. posterior (blue) densities from the Neuhauser (1980) fit. Only σ_L and κ show substantial contraction; rate parameters are prior-driven (cf. Table 9).

Table 14: Comparison of **BayesianDEB** with existing DEB calibration tools.

Feature	AmP/DEBtool	deBInfer	Custom Stan	BayesianDEB
Bayesian inference	—	Yes	Yes	Yes
Pre-built DEB models	Yes	—	—	Yes
Hierarchical models	—	—	Manual	Built-in
DEBtox (TKTD)	Partial	—	Manual	Built-in
Observation model choice	—	—	Manual	Switchable
Prior specification API	—	Basic	Manual	6 families
Temperature correction	Manual	—	Manual	Built-in
Convergence diagnostics	Minimal	Basic	Manual	Automated
PPC	—	—	Manual	Built-in
LOO-CV	—	—	Manual	Built-in
Within-chain parallelism	—	—	Manual	<code>reduce_sum</code>
R package	MATLAB+R	Yes	—	Yes

has no reusable API. **BayesianDEB** combines the model coverage of AmP/DEBtool with the Bayesian inference capabilities of Stan, wrapped in a user-friendly R interface.

Quantitative comparison with deBInfer. To provide a concrete benchmark beyond Table 14, we fitted the simulated individual growth dataset (individual 5 from `eisenia_growth`, 13 weekly observations) with both **BayesianDEB** and **deBInfer** (Boersch-Supan *et al.* 2017)

using matched priors. The **deBInfer** implementation required writing the DEB ODE, likelihood, and prior specification manually (~ 80 lines of R code vs. 6 lines for **BayesianDEB**). The replication script is provided in the supplementary materials (`comparison_debinfer.R`). **deBInfer** uses an adaptive Metropolis-within-Gibbs sampler, producing ~ 100 – 200 effective samples per minute, while the HMC sampler in **BayesianDEB** yields ~ 500 – 1500 effective samples per minute for the same model, reflecting the efficiency advantage of gradient-based sampling for correlated posteriors (Betancourt and Girolami 2015). Posterior medians agreed within Monte Carlo error, confirming that both implementations target the same posterior. Beyond sampling efficiency, the main advantage is the declarative interface: switching observation families, priors, or temperature correction requires changing a single function argument rather than rewriting the model code.

Comparison with DEBtool_M point estimates. For the Neuhauser (1980) dataset (Section 7.1), we compared the **BayesianDEB** posterior medians with the AmP parameter ranges for *E. fetida* (Table 12). The posterior medians for $\{p_{Am}\}$, $[p_M]$, κ , and L_∞ fall within the AmP ranges, while v does not (discussed in Section 7.1). This agreement is expected: the Bayesian posterior with informative AmP-calibrated priors should bracket the deterministic optimum. The key difference is that **BayesianDEB** provides the full posterior distribution, revealing which parameters are constrained by the data (high c) and which merely reflect the prior (low c), whereas **DEBtool_M** returns only point estimates with no formal uncertainty. A direct call-for-call comparison is not feasible because **DEBtool_M** requires MATLAB and uses a different parametrisation and optimisation algorithm; the AmP published ranges serve as the reference standard instead.

10. Discussion

The primary contribution of **BayesianDEB** is not a new statistical method but a *domain-specific bridge* between two mature ecosystems: DEB theory, with its extensive parameter database (AmP; Marques *et al.* 2018) and mechanistic ODE framework (Kooijman 2010; Nisbet *et al.* 2000), and the Bayesian modelling workflow supported by Stan (Carpenter *et al.* 2017; Gabry *et al.* 2019).

Implications for the DEB community. The AmP estimation tool has calibrated DEB parameters for over 3000 species using deterministic optimisation. **BayesianDEB** complements rather than replaces this infrastructure: AmP point estimates serve as informative priors (via `prior_species()`), while the Bayesian posterior propagates their uncertainty to derived quantities and regulatory endpoints such as EC_{50} . This is particularly valuable in ecotoxicology, where risk assessors require not only best estimates but also credible intervals (Billoir *et al.* 2008; Jager *et al.* 2011).

The identifiability message. The structural non-identifiability of DEB rate parameters from growth-only data (Section 6.2) is well known theoretically but rarely quantified. The contraction ratio c provides a practical, parameter-level diagnostic for this. We hope it encourages the DEB community to routinely report prior sensitivity alongside parameter estimates, regardless of the calibration tool used.

Practical guidance for users. Based on the validation results and case studies, we offer three practical recommendations. (i) Always use species-specific priors from `prior_species()` or manually calibrated to the AmP collection; default priors are suitable only for exploratory analyses. (ii) When individual-level data are available for multiple organisms, prefer the hierarchical model: partial pooling substantially improves identifiability of shared parameters (Section 5.3). (iii) Report the contraction ratio c (Section 6.2) alongside posterior summaries and always refit with at least one alternative prior specification to verify robustness of scientific conclusions.

Deterministic process model. **BayesianDEB** uses a deterministic ODE with stochastic observation noise; all unexplained variation is absorbed into σ_L (or, in the hierarchical model, partially separated into $\sigma_{\log\{p_{Am}\}}$). SDE-based process noise would be inferentially richer but substantially more expensive and not natively supported by Stan's ODE solvers. The deterministic formulation matches standard DEB practice and is adequate for controlled laboratory experiments; extension to SDEs is a natural direction for field data.

Sensitivity to modelling choices. The observation model choice can affect inference; we recommend comparing via LOO-CV (Section 5.1). Similarly, T_A misspecification propagates into rate parameter posteriors; when uncertain, a sensitivity analysis over species-specific T_A values should be conducted.

Broader context. **BayesianDEB** joins a growing family of domain-specific Stan wrappers, including **brms** (Bürkner 2017) for regression, **deBInfer** (Boersch-Supan *et al.* 2017) for general ODEs, and various pharmacokinetic packages, that lower the barrier to principled Bayesian modelling in applied sciences. The ODE-in-Stan architecture used here is directly transferable to other mechanistic ecological models (e.g., Lotka-Volterra, SIR epidemiological models), and the SBC validation protocol (Section 6.1) provides a template for rigorous algorithmic testing of any such wrapper package.

11. Current limitations and future work

- The "individual" and "growth_repro" models support exactly one individual; the "debtox" model fits one ODE per concentration group. Hierarchical DEBtox (individual-level) is planned.
- Random effects are on $\{p_{Am}\}$ only; multivariate random effects are planned. Temperature correction is global; time-varying T is planned. Survival and weight endpoints are not yet implemented.
- R-side simulators use **deSolve** LSODA, verified to agree with Stan's BDF solver to $< 10^{-5}$ cm.
- Strong posterior correlations are inherent to the κ -rule (Section 6.2); informative priors are recommended.

- SBC validation covers the individual (397 valid) and hierarchical (37 valid) models; scripts for the remaining two models are provided.
- *Reparametrisation*. Sampling $(L_\infty, g, k_M, \kappa)$ directly would reduce correlations and allow priors on better-identified quantities. This requires per-model Jacobian adjustments and is planned for the next major version.
- *Multi-endpoint inference*. Adding respiration data would independently constrain $[p_M]V$ and substantially reduce prior-dependence of rate parameters; this is a high-priority extension.

12. Summary

BayesianDEB provides the first general-purpose R package for Bayesian Dynamic Energy Budget modelling. By embedding DEB ODEs within Stan’s HMC framework, it delivers full posterior uncertainty propagation to all parameters and derived quantities, principled prior incorporation from the AmP database, hierarchical partial pooling across individuals, and ecotoxicological EC_{50} /NEC estimation with posterior uncertainty. The declarative R interface requires no Stan programming from the user, while threading via `reduce_sum` makes hierarchical and DEBtox models tractable on modern multi-core hardware.

We validated the individual growth model via simulation-based calibration (397 valid replications out of 500; no parameter rejects uniformity at $\alpha = 0.05$) and provided preliminary SBC evidence for the hierarchical model. The identifiability analysis demonstrates that DEB rate parameters are largely prior-shaped under single-individual growth data, reinforcing the need for informative priors or hierarchical pooling. We view this transparency as a feature, not a limitation, of the Bayesian approach: the contraction ratio c and prior sensitivity tools make explicit what deterministic calibration tools obscure behind point estimates.

The real-data case studies confirm that the package produces biologically plausible results on experimental data, and the DEBtox analysis recovers EC_{50} and NEC values consistent with published literature. Future development priorities include time-varying temperature, survival endpoints, multivariate random effects, and reparametrisation in terms of better-identified derived quantities.

Computational details

The results in this paper were obtained using R 4.5.2 (2025-10-31) on `x86_64-pc-linux-gnu` (Ubuntu 22.04) with the packages listed below. **BayesianDEB** 0.1.3 is distributed under the MIT license. The package is available from the SCIOm r-universe at <https://sciom.r-universe.dev/BayesianDEB> and from GitHub at <https://github.com/sciom/BayesianDEB>.

CRAN compatibility and cmdstanr. `cmdstanr` is listed in `Suggests` (not `Imports`), so **BayesianDEB** installs and passes R CMD check on CRAN without requiring `CmdStan`. All functions except `bdeb_fit()` work without Stan: data preparation, prior specification, standalone ODE simulation (`deb_simulate`, `debttox_simulate`), prior predictive checks, and energy flux calculations are available immediately. When a user calls `bdeb_fit()` without

cmdstanr installed, an informative error message with installation instructions is returned. The `Additional_repositories` field in `DESCRIPTION` points to the Stan r-universe (<https://stan-dev.r-universe.dev>), enabling `install.packages("cmdstanr")` to resolve automatically. The package passes R CMD check with 0 errors and 0 warnings; a CRAN submission is planned following the review process.

Replication materials. All scripts and data needed to reproduce the tables and figures in this paper are included in the `MANUSCRIPT_JSS/` directory of the GitHub repository (https://github.com/sciom/BayesianDEB/tree/main/MANUSCRIPT_JSS). The main replication script `code.R` reproduces all non-SBC results (~3 hours). The SBC scripts `sbc_validation.R` (individual, ~90 min), `sbc_hierarchical.R` (hierarchical, ~5 hours), `sbc_growth_repro.R` (growth-reproduction, ~3–5 hours), and `sbc_debtox.R` (DEBtox, ~6–12 hours) are provided separately due to their long runtimes. Pre-computed SBC results are archived in `.rds` files. The **deBInfer** comparison script is `comparison_debinfer.R`.

```
R> install.packages("BayesianDEB",
+   repos = c("https://sciom.r-universe.dev", getOption("repos")))
R> install.packages("cmdstanr",
+   repos = c("https://stan-dev.r-universe.dev", getOption("repos")))
R> cmdstanr::install_cmdstan()
```

R version 4.5.2 (2025-10-31), x86_64-pc-linux-gnu

Attached packages:

BayesianDEB 0.1.3, posterior 1.6.1, bayesplot 1.14.0,
ggplot2 4.0.1, deSolve 1.42, loo 2.8.0

Loaded via namespace:

cmdstanr 0.8.0 (CmdStan 2.36.0), rlang 1.1.7,
cli 3.6.5, Rcpp 1.1.0

Acknowledgments

This work was supported by the Croatian Science Foundation (HRZZ) under the project FUNECOSoil (Functional Ecological Characterization of Soils: The Basis for Ecotoxicological Classification, grant no. IP-2022-10-7177). We thank the Stan development team (Carpenter *et al.* 2017) for the probabilistic programming language and the developers of **cmdstanr** (Gabry and Češnovar 2024), **posterior**, **loo** (Vehtari *et al.* 2017), **bayesplot** (Gabry *et al.* 2019), **ggplot2** (Wickham 2016), and **deSolve** (Soetaert *et al.* 2010) for the R infrastructure on which **BayesianDEB** depends.

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